

Question number	Answer	Mark
7(a)(i)	The only correct answer is B. <i>A is incorrect because nuclei, Golgi apparatus and mitochondria are organelles surrounded by membrane</i> <i>C is incorrect because nuclei, Golgi apparatus and mitochondria are organelles surrounded by membrane</i> <i>D is incorrect because nuclei, Golgi apparatus and mitochondria are organelles surrounded by membrane</i>	(1)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	An explanation that includes the following points: - it is not a plant because it has glycogen granules (1) - it is not an animal because it has a cell wall (1) - it is not a bacterium because it has {nuclei / Golgi apparatus / mitochondria / membrane-bound organelles} (1)	ACCEPT does not have cellulose cell wall IGNORE chloroplast / vacuole IGNORE flagellum / pili / capsule / ER DO NOT ACCEPT ribosomes / cytoplasm / glycogen granules / cell membrane / cell wall unless qualified as {chitin / not peptidoglycan}	(3)

Question number	Answer	Additional guidance	Mark
7(b)	An explanation that includes four of the following points: - there is a correlation between the number of prescriptions and the percentage of resistant <i>E.coli</i> (1) - the use of aminopenicillin acts as a selection pressure (1) - therefore the resistant bacteria reproduce and the non-resistant bacteria die (1) - percentage of resistant <i>E. coli</i> falls when prescriptions fall because non-resistant <i>E. coli</i> are not destroyed (1) - credit a comment about competition between resistant and non-resistant bacteria (1)	ACCEPT pattern / trend IGNORE directly proportional ACCEPT therefore the resistant bacteria {are more likely to reproduce / reproduce more} ACCEPT as the prescriptions go up the number of resistant bacteria go up and when the prescriptions go down the number of bacteria go down for 1 mark if no other marks awarded	(4)

Question number	Answer	Additional guidance	Mark
7(c)	An explanation that includes three of the following points: - because the codes of practice (regarding the prescription of antibiotics) are being ignored (1) - use of antibiotics is a selection pressure (1) - therefore the number of antibiotic resistant bacteria is increasing (1) - our (current) antibiotics may become useless and people will {remain ill / die} (1)	ACCEPT (medical) advice ACCEPT reference to evolutionary race in an appropriate context natural bacterial flora destroyed by antibiotics	(3)

